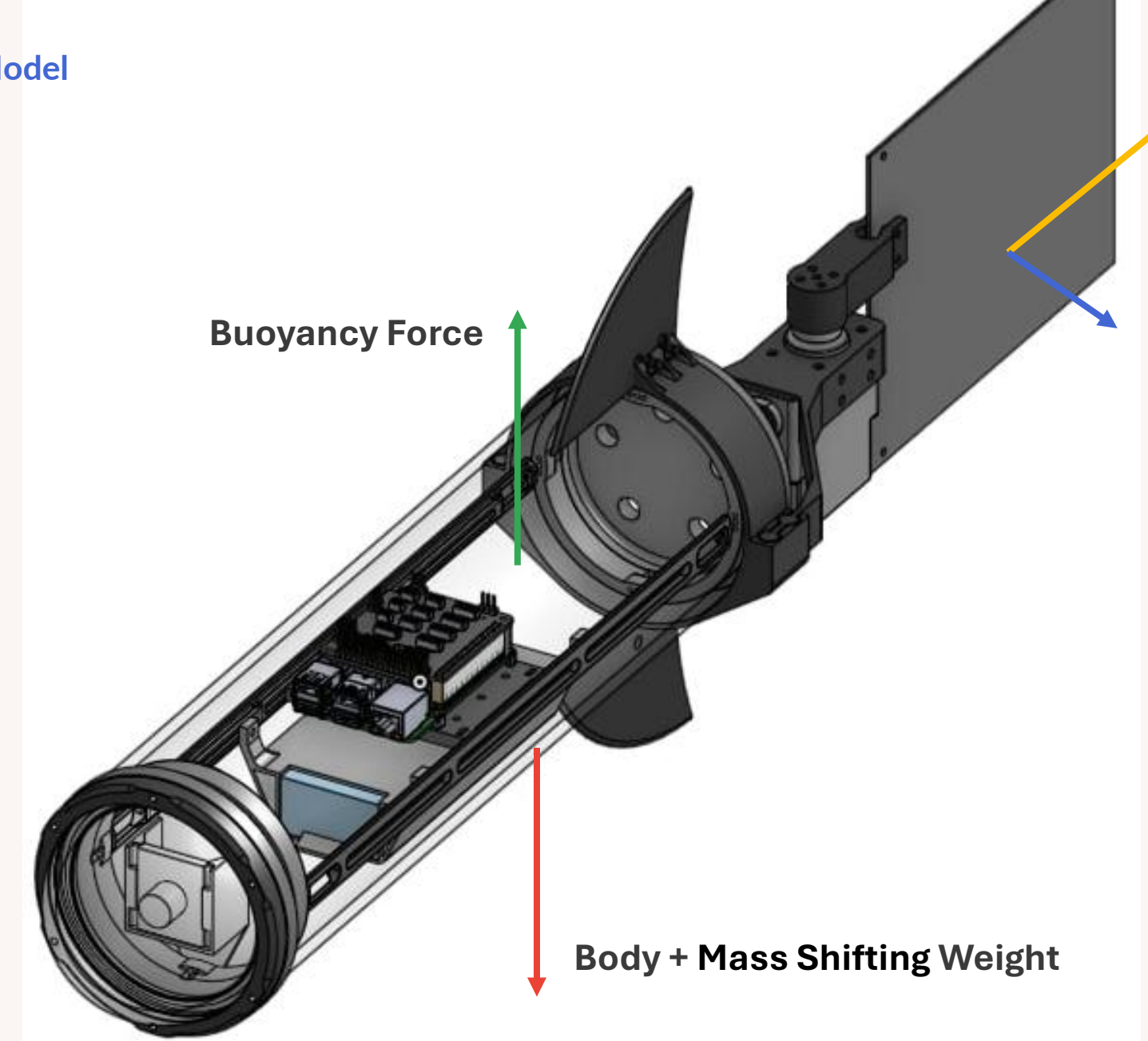
Project Presentation – 3D Simulation

2027 - System Engineering

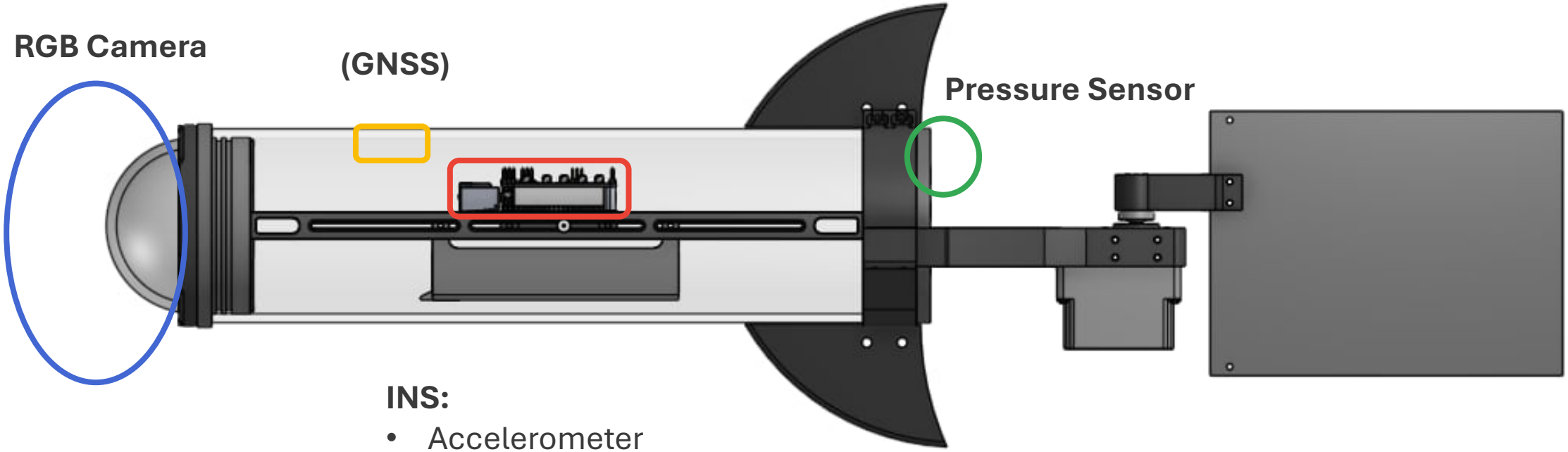
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Physical Model



+ Linear and angular Hydrodynamic drags

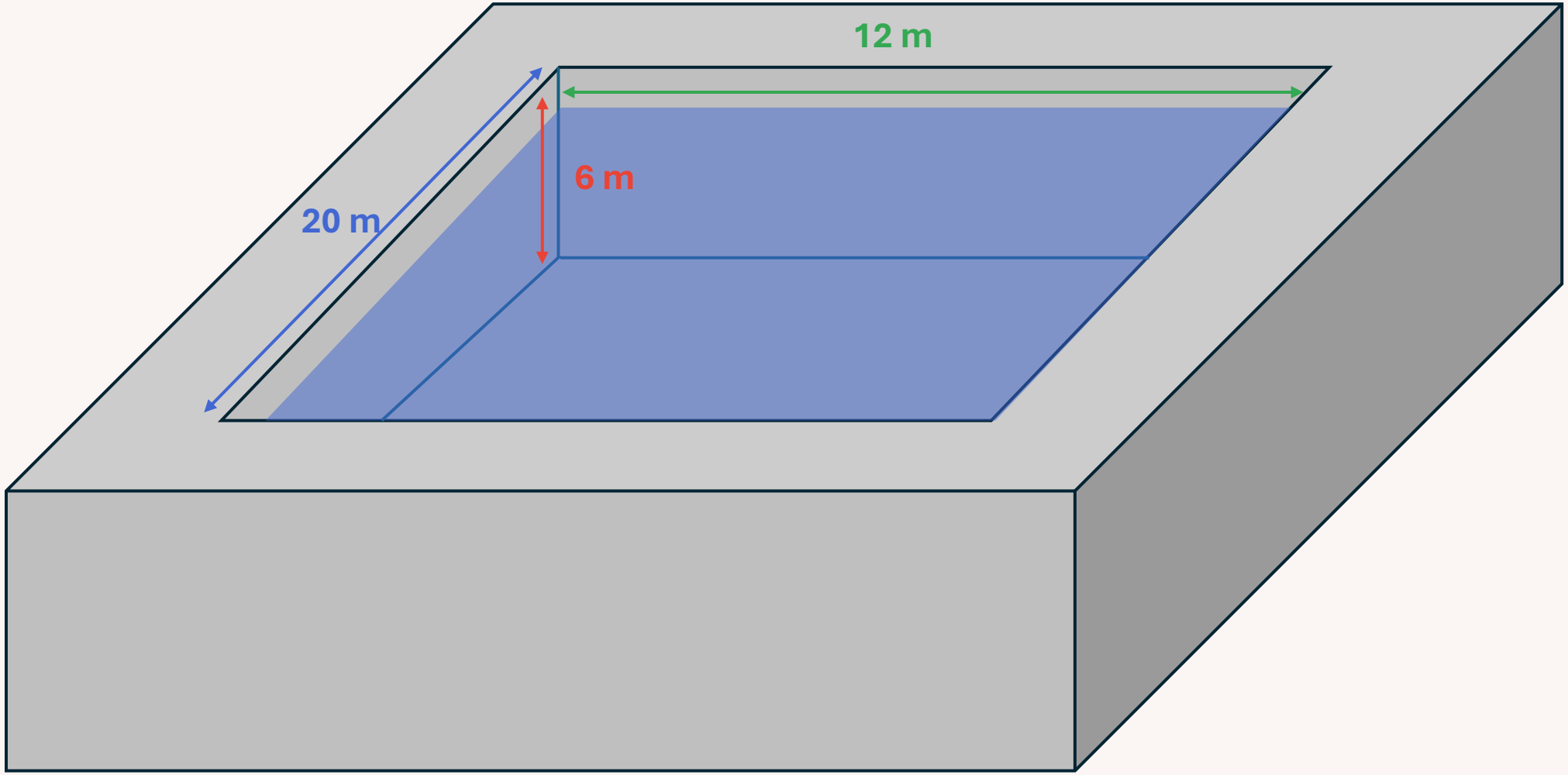
Sensors



INS:

- Accelerometer
- Gyroscope
- Magnetometer

3D environment



Objectives

Physics
- Weight,
- Buoyancy force,
- Fin lift,
- Fin thrust,
- + linear and angular drags.

Sensors
- RGB camera,
- INS : accelerometer, magnetometer, gyroscopes,
- Pressure sensor,
- (GNSS)

3D model
- On Ubuntu 24.04, you can use free, open source online software Onshape

3D scene
- Terrain
- Environment
- (Skybox, props ...)

Then, add high-level intelligence to have heading following, depth regulation etc..
- Navigation
- Guidance
- Control

Thank you for your attention

Do you have any question?

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